A More details for workload generation in Section 4.1

A.1 More details for Example 4.1

The full details of the query shown in Example 4.1 are presented as follows:

SELECT c.c_custkey, c.c_name, c.c_nation, SUM(lo.lo_revenue) AS total_revenue, AVG(lo.lo_quantity) AS avg_quantity, (SELECT COUNT(*) FROM lineorder lo2 WHERE lo2.lo_custkey = c.c_custkey AND lo2.lo_orderdate >= 19921540) AS recent_orders_count FROM customer c JOIN lineorder lo ON c.c_custkey = lo.lo_custkey JOIN dim_date dd ON lo.lo_orderdate = dd.d_datekey WHERE dd.d_year >= 1998 AND lo.lo_discount BETWEEN 5 AND 8 GROUP BY c.c_custkey, c.c_name, c.c_nation HAVING SUM(lo.lo_revenue) > 959691 ORDER BY total_revenue DESC LIMIT 100;

A.2 More details for Example 4.2

In addition to the query shown in Example 4.1, the additional three queries used to construct the workload, $\mathcal{W}_{0,0}$ are listed as follows:

(1) SELECT c.c_custkey, c.c_name, c.c_nation, s.s_name, s.s_nation, p.p_brand1, p.p_category, dd.d_year, dd.d_sellingseason, COUNT(DISTINCT lo.lo_orderkey) AS order_count, SUM(lo.lo_quantity) AS total_quantity, SUM(lo.lo_extendedprice) AS total_extended_price, SUM(lo.lo_revenue) AS total_revenue, AVG(lo.lo_discount) AS avg_discount, AVG(lo.lo_tax) AS avg_tax, MIN(lo.lo_orderdate) AS first_order_date, MAX(lo.lo_orderdate) AS last_order_date, COUNT(*) FILTER (WHERE lo.lo_orderpriority = '1-URGENT') AS urgent_orders, COUNT(*) FILTER (WHERE lo.lo_shipmode IN ('AIR', 'REG AIR')) AS air_shipments FROM lineorder lo JOIN customer c ON lo.lo_custkey = c.c_custkey JOIN supplier s ON lo.lo_suppkey = s.s_suppkey JOIN part p ON lo.lo_partkey = p.p_partkey JOIN dim_date dd ON lo.lo_orderdate = dd.d_datekey WHERE lo.lo_quantity > 24 AND lo.lo_extendedprice > 38010 AND lo.lo_orderdate BETWEEN 19970971 AND 19953889 AND p.p_size BETWEEN 49 AND 34 AND dd.d_year >= 1997 AND (c.c_mktsegment = 'AUTOMOBILE' OR s.s_region = 'AFRICA') GROUP BY c.c_custkey, c.c_name, c.c_nation, s.s_name, s.s_nation, p.p_brand1, p.p_category, dd.d_year, dd.d_sellingseason HAVING SUM(lo.lo_revenue) > 469233 AND COUNT(DISTINCT lo.lo_orderkey) >= 77 AND AVG(lo.lo_discount) < 4 ORDER BY total_revenue DESC, order_count DESC, last_order_date DESC LIMIT 276

(2) SELECT c.c_custkey, c.c_name, c.c_nation, s.s_nation, p.p_category, p.p_brand1, d.d_year, d.d_sellingseason, SUM(lo.lo_revenue) AS total_revenue, SUM(lo.lo_quantity) AS total_quantity, AVG(lo.lo_discount) AS avg_discount, COUNT(*) AS order_count, RANK() OVER (PARTITION BY c.c_region, s.s_region ORDER BY SUM(lo.lo_revenue) DESC) AS revenue_rank_in_region, LAG(SUM(lo.lo_revenue), 1) OVER (PARTITION BY c.c_custkey, p.p_category ORDER BY d.d_year) AS prev_year_revenue FROM lineorder lo JOIN customer c ON lo.lo_custkey = c.c_custkey JOIN supplier s ON lo.lo_suppkey = s.s_suppkey JOIN part p ON lo.lo_partkey = p.p_partkey JOIN dim_date d ON lo.lo_orderdate = d.d_datekey WHERE lo.lo_orderdate BETWEEN 19945854 AND 19934330 AND lo.lo_quantity > 36 AND lo.lo_discount BETWEEN 6 AND 0 AND p.p_size >= 41 AND d.d_year >= 1993 GROUP BY c.c_custkey, c.c_name, c.c_nation, s.s_nation, p.p_category, p.p_brand1, d.d_year, d.d_sellingseason, c.c_region, s.s_region HAVING SUM(lo.lo_revenue) > 2857488 AND COUNT(*) > 673 ORDER BY total_revenue DESC, d.d_year, c.c_nation, s.s_nation LIMIT 7038

(3) SELECT c.c_custkey, c.c_name, c.c_nation, s.s_nation, SUM(lo.lo_revenue) AS total_revenue, AVG(lo.lo_quantity) AS avg_quantity, COUNT(*) AS order_count FROM lineorder lo JOIN customer c ON lo.lo_custkey = c.c_custkey JOIN supplier s ON lo.lo_suppkey = s.s_suppkey JOIN dim_date d ON lo.lo_orderdate = d.d_datekey WHERE lo.lo_discount BETWEEN 10 AND 13 AND d.d_year = 1992 AND lo.lo_quantity > 50 GROUP BY c.c_custkey, c.c_name, c.c_nation, s.s_nation HAVING SUM(lo.lo_revenue) > 7756533 ORDER BY total_revenue DESC LIMIT 100